Exact homogeneous-level projection norms on Boolean cubes

Automated mathematical discovery run `fa_banach_001`

13 August 2026

Status. Candidate full solution, likely valid, subject to human review. The result answers the source's Section 4.3 question affirmatively and proves the stronger exact formula for every fixed degree and Fourier level.

1 Source question

The source is A. Defant, M. Mastyło, and A. Pérez, *On the Fourier spectrum of functions on Boolean cubes*, arXiv:1706.03670 [1]. On source p. 24, Section 4.3 asks whether $1+\sqrt{2}$ is the best constant C such that

$$\|f_m\|_\infty \le C^d \|f\|_\infty$$

for every real $f : \{-1,1\}^n \to \mathbb{R}$ of Fourier degree at most d and every $0 \le m \le d$.

24 A. DEFANT, M. MASTYŁO AND A. PÉREZ

4.3. **Norm estimates for homogeneous parts.** In view of Lemma 1.3,(4) and the difficulties described at the beginning of section 3 it would be interesting to know whether $C = 1+\sqrt{2}$ is the best constant satisfying

$$\|f_m\|_\infty \le C^d \|f\|_\infty$$

for every $f : \{\pm 1\}^n \to \mathbb{R}$ of degree d and $0 \le m \le d$. See also the commend made at the end of section 3.

Write the Fourier–Walsh polynomial of f as

$$F(x) = \sum_{S\subseteq[n]} \widehat{f}(S)x^S, \qquad x^S = \prod_{j\in S} x_j,$$

and let $F_m = f_m = \sum_{|S|=m} \widehat{f}(S)x^S$. The polynomial F is multi-affine, and therefore

$$\|F\|_{[-1,1]^n} = \|f\|_{\{-1,1\}^n}\,. \tag{1}$$

For $0 \le m \le d$, define the classical coefficient constant

$$M_{m,d} := \sup\left\{ |[t^m]p| : \deg p \le d,\ \sup_{-1\le t\le 1} |p(t)| \le 1 \right\}. \tag{2}$$

Markov's sharp coefficient theorem says that $M_{m,d}$ is the absolute t^m coefficient of T_d when $m \equiv d \pmod 2$, and of T_{d-1} otherwise [1, Eqns. (12)–(13)]; see also [2, Ch. 3].

2 Proof intuition

The known upper bound reduces the multivariable question to one variable: on each radial line, $F(tx)$ is a degree-d polynomial whose t^m coefficient is $F_m(x)$, so Markov's theorem gives the constant $M_{m,d}$. What was missing is sharpness inside the multi-affine class.

To recover a one-variable Chebyshev extremizer without repeated variables, spread its single variable over more and more Boolean coordinates:

$$f_n(\varepsilon) = T_{d'}\left(\frac{\varepsilon_1 + \cdots + \varepsilon_n}{n}\right).$$

This remains bounded by one. After reducing powers by $x_j^2 = 1$, the degree-m part of the mth power of the average is asymptotically the ordinary squarefree term: collisions among indices have lower order in n. All higher Chebyshev powers contribute only collision terms and vanish in the limit. The level-m value therefore tends exactly to the extremal Chebyshev coefficient.

3 Exact result

For fixed m, d, put

$$\Lambda_{m,d} := sup_{n\geq 1} \sup_{\substack{0\neq f:\{-1,1\}^n\to\mathbb{R} \\ \deg f\leq d}} \frac{\|f_m\|_\infty}{\|f\|_\infty}.$$

Theorem 1. *For every $d \geq 1$ and $0 \leq m \leq d$,*

$$\boxed{\Lambda_{m,d} = M_{m,d}.}$$

Consequently, the smallest C for which $\|f_m\|_\infty \leq C^d \|f\|_\infty$ holds uniformly in n, d, m is

$$\boxed{C = 1 + \sqrt{2}.}$$

Proof. We first prove the upper bound. Let F be the multi-affine polynomial of a function f of degree at most d. For $x \in [-1,1]^n$, set

$$p_x(t) := F(tx) = \sum_{r=0}^{d} t^r F_r(x).$$

If $t \in [-1,1]$, then $tx \in [-1,1]^n$. Hence $\|p_x\|_{[-1,1]} \leq \|F\|_{[-1,1]^n}$, while the coefficient of t^m in p_x is $F_m(x)$. The definition (2), followed by (1), gives

$$|F_m(x)| \leq M_{m,d} \|F\|_{[-1,1]^n} = M_{m,d} \|f\|_{\{-1,1\}^n}.$$

Taking the supremum over x proves $\Lambda_{m,d} \leq M_{m,d}$.

For the reverse inequality, choose

$$d' = \begin{cases} d, & m \equiv d \pmod 2, \\ d-1, & m \not\equiv d \pmod 2, \end{cases} \qquad T_{d'}(t) = \sum_{k=0}^{d'} a_k t^k.$$

Then $|a_m| = M_{m,d}$. For each n, define

$$f_n(\varepsilon) = T_{d'}\left(\frac{\varepsilon_1 + \cdots + \varepsilon_n}{n}\right), \qquad \varepsilon \in \{-1,1\}^n. \tag{3}$$

The argument of $T_{d'}$ lies in $[-1,1]$, so $\|f_n\|_\infty \leq 1$. Let Q_n be the unique multi-affine polynomial agreeing with (3) on $\{-1,1\}^n$; equivalently, reduce every x_j^r using $x_j^2 = 1$. Thus Q_n is precisely the Fourier–Walsh polynomial of f_n and has degree at most d'.

For integers $k, m \geq 0$, let $N_{n,k,m}$ be the number of ordered k-tuples $(i_1, \ldots, i_k) \in [n]^k$ for which exactly m indices occur an odd number of times. Expanding $(x_1 + \cdots + x_n)^k$ and reducing modulo $x_j^2 - 1$ shows that

$$(Q_n)_m(\mathbf{1}) = \sum_{k=0}^{d'} a_k n^{-k} N_{n,k,m}. \tag{4}$$

If $k < m$, then $N_{n,k,m} = 0$. If $k = m$, each of the m occurring indices must occur once, and hence

$$N_{n,m,m} = (n)_m = n(n-1)\cdots(n-m+1). \tag{5}$$

Finally suppose $k > m$. If such a tuple uses r distinct indices, m of them have odd positive multiplicity and the other $r - m$ have even multiplicity at least two. Therefore

$$k \geq m + 2(r - m), \qquad r \leq \frac{k+m}{2}.$$

For fixed k, m, choosing the distinct indices and their multiplicities gives the uniform estimate

$$N_{n,k,m} = O_{k,m}\Big(n^{(k+m)/2}\Big). \tag{6}$$

Equations (4)–(6) now imply

$$(Q_n)_m(\mathbf{1}) = a_m \frac{(n)_m}{n^m} + \sum_{k=m+1}^{d'} a_k O_{k,m}\Big(n^{-(k-m)/2}\Big) \longrightarrow a_m.$$

Since $\|f_n\|_\infty \leq 1$,

$$\Lambda_{m,d} \geq \limsup_{n\to\infty} \|(f_n)_m\|_\infty \geq \lim_{n\to\infty} |(Q_n)_m(\mathbf{1})| = |a_m| = M_{m,d}.$$

This proves the exact identity.

If "degree d" is read as *exactly* d rather than at most d, the only extra case to note is $d' = d - 1$ (where necessarily $m < d$). On d fresh Boolean coordinates η, replace f_n by

$$f_n(\varepsilon) + \delta\eta_1 \cdots \eta_d.$$

Its degree is exactly d, its level-m part is unchanged, and its norm is at most $1 + \delta$. Letting $\delta \downarrow 0$ gives the same lower bound.

It remains to identify the best common exponential base. Put $A = 1 + \sqrt{2}$, so $A^{-1} = \sqrt{2} - 1$. The nonzero coefficients of T_r alternate in sign along their common parity. Consequently all terms have the same phase at i, and the standard closed form for T_r gives

$$\sum_{j=0}^{r} |[t^j]T_r| = |T_r(i)| = \frac{A^r + (-1)^r A^{-r}}{2}. \tag{7}$$

In particular every coefficient of T_r is at most A^r. The same bound for T_d and T_{d-1} gives $M_{m,d} \leq A^d$, so $C = A$ works. Conversely, (7) and the fact that T_r has at most $r + 1$ coefficients show that some coefficient has magnitude at least $(A^r - A^{-r})/(2(r + 1))$. Its rth root tends to A. Thus no smaller uniform base can work. This completes the proof. □

4 Checks, scope, and novelty audit

The proof is exact and has no computational dependency. As a sanity check, the symmetric Fourier formula was evaluated for $(d, m) = (6, 2), (7, 3), (10, 6)$; the level values converge to the predicted Chebyshev coefficients $18, 56, 1120$. The script in `code/check_chebyshev_limits.py` reproduces this check.

The result answers only Section 4.3. It does not resolve the distinct Aaronson–Ambainis conjecture or the source's question about subexponential Lorentz Bohnenblust–Hille constants.

A bounded search on 13 August 2026 covered the arXiv identifier and title, the exact Section 4.3 wording, and combinations of "homogeneous parts", "Boolean cube", "Fourier level projection", "Markov coefficient", and "Chebyshev". It found later general dimension-free bounded-projection work, including [3], but no exact statement of $\Lambda_{m,d} = M_{m,d}$ and no explicit resolution of Section 4.3. Novelty confidence is therefore provisional. Human review should focus on the Fourier/multilinearization identification and the count in (6).

References

[1] A. Defant, M. Mastyło, and A. Pérez, *On the Fourier spectrum of functions on Boolean cubes*, Math. Ann. 374 (2019), 653–680; arXiv:1706.03670.

[2] P. Borwein and T. Erdélyi, *Polynomials and Polynomial Inequalities*, Graduate Texts in Mathematics 161, Springer, 1995.

[3] L. Becker, O. Klein, J. Slote, A. Volberg, and H. Zhang, *Dimension-free discretizations of the uniform norm by small product sets*, Invent. Math. 239 (2025), 469–503; arXiv:2310.07926.